Scoped O.1 on the Cone, v2: A Structure Theorem for the Apex Inventory, Ward Closure at Two and Three Fields, and the One-Loop Apex Poles of the One-Antifield Row

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Abstract

We continue the transport of the scoped closure O.1 of the Enter companion programme (XVI-Q record Q11: one loop, order κ^2, bulk two-point sector of $S_{\rm inv}$, $m = 0$, frozen de Donder gauge, XVI-P prescription with the p22 evanescent counterterms) to the flat cone $C_\alpha \times \mathbb{R}^2$. This version is self-contained and supersedes v1. It keeps all v1 results, which were re-run and survive (one engine correction is disclosed in §8), and adds the following. (A) A *structure theorem*: on the curvature-regular (CR) branch the apex counterterm inventory is α-independent away from the resonant set, which is exactly $\{1/3, 1/2, 2/3\}$. Every resonant point was run on both branches with two primes. The flat sanity count 5/2/3/1 is reproduced inside the apex engine. (B) A conditional theorem: if the cone regularisation is BRST-invariant up to evanescent terms, the 26 Ward-constrained apex combinations have *vanishing* one-loop pole coefficients. The remaining 8 closed directions (6 exact, 2 scalar) stay open. (C,D) The one-antifield ($E = 1$) row, which vanishes in flat space, is now computed on the cone. The three one-loop tadpoles D14a,b,c have apex poles with exact rational α-dependence. The scalar loop reproduces Fursaev's a_2^Σ exactly, both coefficients including normalisation. The graviton and ghost loops are *separately* not ST-closed, but their sum on CR is closed exactly. In units of the scalar normalisation, graviton+ghost (CR) equals 2×(scalar) plus $-\frac{4\pi}{9}\frac{(2\alpha-1)(7\alpha-4)}{\alpha}\,\bar{R} - \frac{4\pi}{9}\frac{(2\alpha-1)(4\alpha-1)}{\alpha}(\bar{R}_{ii} - 2\bar{R}_{ijij})$. This has a nonzero flat limit, $-\frac{4\pi}{3}$ for each coefficient, the analogue of the Fursaev–Miele spin-2 remnant. On the Friedrichs branch the same sum is *not* closed, by $\propto (\alpha-1)(2\alpha-1)(3\alpha-1)$; this is the one-loop face of the v1 domain obstruction. (E) The three-field apex cohomology at ghost number one on CR, at generic α, is $H^1_{(3)} = 0$ (83 closed = 83 exact out of 194; two primes, two values of α, $\sigma_0^2 = 0$ checked). (F) Independent second implementations: mode sums against the closed Bernoulli form, for the Synge coincidence data (error $O(\lambda^3)$) and for the spin-2 Ward residuals (agreement to all printed digits). The two-point bubble graphs D01, D03, D04, D06, D11 and D13 on the cone remain open, with a precisely stated obstruction. **O.1 on the cone remains NOT_CLOSED.**

1 Background, scope and conventions

Enter background. The Enter companion programme (repository `cheungngo/enter-companion`) is a sequence of machine-checked landmarks about perturbative gravity coupled to a scalar on fixed backgrounds. XVI-P fixes the prescription: DRED with quasi-d-dimensional loop momenta, quasi-4-dimensional vertex indices, $\overline{\rm MS}$, and the p22 evanescent-counterterm contract. XVI-Q certifies the scoped closure O.1 (Q03/Q11). XVI-R imports Q11. All three trees are read-only inputs here; we used copies in `/tmp/o1cone_ro`, and the

tree hashes of the originals agree before and after this work (§10). The cone and its CR extension, the I-B-G-S v2 spin-2 Green operators, are *inputs*, not consequences. A separation is not a bridge.

Scope. *O.1 on the cone* is the Q11 statement with Minkowski space replaced by $M_\alpha = \{-dt^2 + dr^2 + \alpha^2 r^2 d\varphi^2 + dz^2\}$, glued by $-\mu$, $\mu = 2\pi(1-\alpha)$, $t := 1/\alpha > 1$. Sector, order, prescription and gauge are unchanged. Spin-2 fields obey the CR condition of I-B-G-S v2. Friedrichs (F) is used only for comparison. `bulk_symmetry_restored = false`.

Kernels. Write $\epsilon = \theta' - \theta$ for the developed angle, $\psi = \epsilon/\alpha$, $\cosh\eta = (r^2 + r'^2 + \Delta_\parallel^2)/(2rr')$ and $u = \eta/\alpha$. The helicity-S label kernels are

$$G_b = \frac{S_b(u,\psi)}{8\pi^2\alpha\, rr' \sinh\eta}, \qquad S_b = f_b(u - i\psi) + f_{1-b}(u + i\psi), \qquad f_b(\zeta) = \frac{e^{-b\zeta}}{1 - e^{-\zeta}},$$

with $b(S) = S\alpha$ for $S \geq 0$ and $b(S) = 1 + S\alpha$ for $S < 0$ on CR, and $b = \mathrm{frac}(S\alpha)$ on F. The flip is $S_b = S_{b-1} + 2e^{i(b-1)\psi}\sinh((b-1)u)$. In Lorentzian signature, $\Delta\tau^2 \to -\Delta t^2$, and the time-ordered function is the continued G_E.

Exact Synge data. Expanding in Bernoulli polynomials gives, exactly,

$$S_b = \sum_{n\geq 0} \frac{B_n(b)}{n!}\big[(-1)^n(u - i\psi)^{n-1} + (u + i\psi)^{n-1}\big],$$

and the subtracted remainder $R_b = G_b - G_{\mathrm{flat}} = \Phi_b/(8\pi^2 rr')$, with $\Phi_b = T(b,\alpha) - T(0,1)$ (T = the $n \geq 2$ tail divided by $\sinh\eta$), is real-analytic at the diagonal of M_{reg}. From $R(x, x+\delta) = A + B\cdot\delta + \delta\cdot C\cdot\delta + O(\delta^3)$ we use the Synge rules $\partial_{x'_i} R \to B_i$, $\partial_{x_i} R \to \partial_i A - B_i$, $\partial_{x'_i}\partial_{x'_j} R \to 2C_{ij}$, $\partial_{x_i}\partial_{x'_j} R \to \partial_i B_j - 2C_{ij}$, and $\partial_{x_i}\partial_{x_j} R \to \partial_i\partial_j A - \partial_i B_j - \partial_j B_i + 2C_{ij}$. The coincidence limit is $a(b) := 4\pi^2\alpha\, r^2 R_b(x,x) = \frac{1}{12}(\alpha^{-1} - \alpha) - \frac{b(1-b)}{2\alpha}$ (v1, reproduced exactly).

2 Results carried over from v1 (re-verified)

Theorem 2.1 (Regular-region locality)**.** *Let $\Gamma_2^{(1)}$ be the one-loop κ^2 two-point functional of the retained sector on M_α, regularised by the XVI-P prescription with the longitudinal directions continued dimensionally. On $C_c^\infty(M_{\mathrm{reg}}^2)$, the pole part of $\Gamma_2^{(1)}$ equals the flat XVI-Q one: the same 21 directions (11 p14r and 10 p22 evanescent) and the same coefficients, h.h#0..4 $= \frac{49i}{480}, \frac{41i}{160}, -\frac{49i}{240}, -\frac{41i}{80}, \frac{43i}{120}$, c.c̄#0,1 $= -\frac{3}{8}, \frac{1}{4}$, c.K#0..2 $= 0, -\frac{3}{8}, \frac{1}{4}$, $\phi\phi$#0 $= 0$. The σ_0 closure and the ST rows hold there with the flat coefficients. This is true on both branches.*

Proof. $S_b(-u,\psi) = -S_b(u,\psi)$, so G_b is even in η, and $G_b - G_{\mathrm{flat}}$ is real-analytic near the diagonal of M_{reg}^2 for $|\psi| < \pi$. Write each propagator as $H + R$. Then $H\cdot R$ and $R\cdot R$ are finite on $C_c^\infty(M_{\mathrm{reg}}^2)$, and the pole is the flat $H\cdot H$ local functional, which is the Q02 one. The vertices and σ_0 are local. □

Theorem 2.2 (CR reproduces Fursaev–Miele and Kabat; v1)**.** *Exact rational arithmetic gives the following. The F vector labels give $2a(0) + a(\alpha) + a(1-\alpha) = 4a_0 - (1-\alpha)$, which is Kabat's contact term and FM eq. (2.11). The CR spin-2 labels give $4a_0 + 2(a(\alpha) + a(1-\alpha)) + a(2\alpha) + a(1-2\alpha) = 10a_0 + 6(\alpha - 1) + 2$ for all $\alpha \in (0,1)$, which is FM eq. (3.47) at $d = 4$ including the $+8\pi$ term. For $\alpha > 1/2$, F misses it by $-2(2 - \alpha^{-1})$.*

The v1 statement that the CR E_0 channels $(\pm 2, \mp 1)$ carry the FM "dipole" term is a coefficient match, not a derivation. Section 5 finds the same channels producing a flat-limit remnant at the next order.

3 (A) The apex inventory: flat count, resonances, structure theorem

Set-up (as in v1). Counterterms are local functionals on $W = \{r = 0\}$, $\int_W d^2y \sum J_1[\Phi_1]J_2[\Phi_2]$. The jets (C, n, m) of the I-B-G-S v2 lattice have rotation charge n and radial power m; dimensions are $[h] = [c] = [\bar{c}] = [\phi] = 1$, $[b] = [K] = [c^*] = 2$. The constraints are power counting ($T = 4$ at gh0, $T = 5$ at gh1, $T = 3$ at gh-1), $n_1 + n_2 = 0$, $O(2)_\parallel$ pairings, φ-reflection, Bose/Grassmann symmetry and ghost number. The linear BRST operator is $sh = K_2 c$, $s\bar{c} = b$, $sK = Eh + C_2^T b$ and $sc^* = K_2^T K - (C_2 K_2)^T \bar{c}$. Ranks are computed modulo $p_1 = 10^9 + 9$ and $p_2 = 998244353$ on random jet configurations. These are Schwartz–Zippel certificates: a rank deficit is reported only if it persists at both primes, and the probability of a false drop is $\leq \deg / p$ per sample.

Proposition 3.1 (A1: flat sanity count inside the apex engine)**.** *Using the same candidate builder and modular rank routine with bulk (flat, W-free) momentum-space functionals, the flat two-point counterterm counts are h.h: 10 raw patterns, rank 5; c.$\bar{c}$: 2/2; c.K: 4/3; $\phi\phi$: 1/1. These are identical at both primes and equal the XVI-P/XVI-Q counts 5/2/3/1 (`logs/v2_flat_count.json`).*

Theorem 3.2 (A3: structure theorem)**.** *Let $\alpha \in (0, 1)$ on CR. (i) Every lattice jet has radial power $m = |n|\, t + A$ with $A \in \mathbb{Z}$ and $A \geq -2$. The value $A = -2$ (at $|n| = 1$) occurs only in the CR-flipped channels $(\pm 2, \mp 1)$, where $m = t - 2 + 2j$. Under F the same channels with $t < 2$ carry $A = -1$ instead. (ii) An apex candidate with integer scaling weight (the only kind that multiplies a DR pole) and some jet with $n \neq 0$ exists only if $Nt \in \mathbb{Z}$ for an $N = \sum_i |n_i| \geq 2$ with $N < K \leq M + 2J$, where M is the dimension budget and J the number of flip-capable legs. For two-point candidates with $T \leq 5$ this happens exactly for $\alpha \in \{1/3, 1/2, 2/3\}$. (iii) At every other α only $n = 0$ jets enter. Their data are α-independent ($a = s$, integer m, operator coefficients $(m \pm a)/2$), so all candidate, closed, exact and cohomology counts are α-independent. This holds for any number of fields.*

Proof. (i) follows from the orders $\nu = |n/\alpha + \sigma|$ with $|\sigma| \leq 2$, together with the CR choice of E_0 in $(\pm 2, \mp 1)$. (ii) Rotation invariance forces $\sum n_i = 0$. Non-integer parts of the m's cancel only if $\sum |n_i|\, t \in \mathbb{Z}$, and the budget gives the bound. We then enumerated all N/K in the window and also scanned the grid $\alpha = k/97$ (`code/resonances.py`, `logs/v2_resonances.json`). CR is non-empty only at $1/3, 1/2, 2/3$. F is non-empty at 56 grid points, the smallest being $1/3$, and at $\alpha \leq 1/2$ only at $\{1/3, 1/2\}$, where F and CR coincide. (iii) For $n = 0$ the jet coefficients involve only σ and integers. □

Corollary 3.3 (A: all α)**.** *For every $\alpha \in (0, 1)$ on CR, $H^0_{(2)} = 2$ (the two $\phi\phi$ terms $\delta_W \phi\, \partial_\parallel^2 \phi$ and $\delta_W \phi\, \partial_r^2 \phi|_{m=2}$) and $H^1_{(2)} = 0$. Off the resonant set this follows from Theorem 3.2(iii) and the generic computation. On the resonant set it follows from the exhaustive runs above. The resonances $\alpha = 1/4$ and $2/5$ asked about are not resonant for $T \leq 5$, and their runs give the generic counts, as predicted.*

Remark 3.4. The corollary is a proof modulo the Schwartz–Zippel nature of modular ranks. The candidate enumeration is exhaustive by construction, and the α-independence is a theorem.

α (CR)	$hh/\bar{c}c/Kc/\phi\phi$	N	Z	B	H^0	gh1 closed/exact, H^1
generic $(3/10, 1/4, 2/5, 7071/10000)$	18/8/6/2	34	8	6	2	26/26, 0
1/3 (new)	19/8/6/2	35	8	6	2	27/27, 0
1/2	24/9/7/2	42	9	7	2	33/33, 0
2/3	19/8/6/2	35	8	6	2	27/27, 0
F, $\alpha = 2/3$ (diagnostic)	22/8/7/2	39	9	6	3	32/30, "2"; 59 foreign
F, $\alpha = 7071/10000$ (diagnostic)	21/8/7/2	38	9	6	3	31/29, "2"; 58 foreign

Table 1: v2 apex engine (jet cap T), both primes identical. At $\alpha \leq 1/2$, F equals CR. The F rows are diagnostic only, because K_2c leaves the F domain ("foreign" samples).

4 (E,F) Ward identities: two fields, three fields, second implementation

Theorem 4.1 (Two-point, CR; v1, re-run)**.** *At every α in the table, gh1 two-point closed = exact, $\sigma_0^2 = 0$, and $H^1_{(2)} = 0$. Every apex-supported breaking of the linearised ST identity that obeys the WZ condition is $\sigma_0\Gamma_W$ with Γ_W in the inventory.*

Three fields (E). We use the minimal BV complex; the pair $(\bar{c}, b)$ is contractible. Its variations are $sh = K_2c$, $sK = Eh$, $sc^* = K_2^T K$, $s\phi^* = \Box\phi$ and $sc = s\phi = 0$. The sectors are as follows.

- gh0, $T = 5$: hhh, $\phi\phi h$, Kch, c^*cc, $\phi^*\phi c$.
- gh1, $T = 6$: hhc, $\phi\phi c$, Kcc.
- gh2, $T = 7$: hcc.

Only $n = 0$ jets enter, which is sufficient at non-resonant α by Theorem 3.2(iii). Identical bosons are polarised with weight $1/n!$, and the third momentum is fixed by $q_3 = -q_1 - q_2$. Identical ghosts are antisymmetrised as commuting placeholders. The relative signs of the maps are fixed by requiring $\sigma_0^2 = 0$, which we check. The maps σ_0 are as follows.

- From gh0 to gh1:
  - $hhh \to hhc$: $3F(K_2c, h, h)$;
  - $\phi\phi h \to \phi\phi c$: $G(\phi, \phi, K_2c)$;
  - $Kch \to hhc$: $\frac{1}{2}[H(Eh_1, c, h_2) + H(Eh_2, c, h_1)]$;
  - $Kch \to Kcc$: $H(K, c_1, K_2c_2)$, antisymmetrised;
  - $c^*cc \to Kcc$: $S(K_2^T K, c_1, c_2)$;
  - $\phi^*\phi c \to \phi\phi c$: the polarisation of $\Phi(\Box\phi_1, \phi_2, c)$.
- From gh1 to gh2:
  - $hhc \to hcc$: $2[D(K_2c_1, h, c_2) - D(K_2c_2, h, c_1)]$;
  - $Kcc \to hcc$: $D(Eh, c_1, c_2)$.

Theorem 4.2 (E: three-field ghost-number-one apex cohomology)**.** *On CR at* $\alpha = 7071/10000$ *with both primes (and at a second generic* $\alpha = 309/500$*, prime* p_1*: identical dimensions and* 83/83*), the gh0 dimensions are hhh 51 (of 780 raw),* $\phi\phi h$ *12,* Kch *31,* c^*cc *4 and* $\phi^*\phi c$ *3. The gh1 dimensions are hhc 161 (of 856),* $\phi\phi c$ *10 and* Kcc *23, for a total of 194. The WZ-closed subspace has dimension* 83*, and* σ_0(gh0) *has dimension* 83*. The images lie in the span, and their coordinates satisfy every closure row, so* $\sigma_0^2 = 0$*. Hence* $H^1_{(3)} = 0$*, and the same holds at every non-resonant* α *by Theorem 3.2(iii).*

Remark 4.3. This is the trilinear, α-generic part of the nonlinear question. It is the linearised-in-σ_0 cohomology of three-field functionals. The full nonlinear descent, in which $S^{(1)}$ mixes (2) $\to$ (3) through the cubic vertices, needs $H^1_{(2)} = 0$ (Theorem 4.1) and $H^1_{(3)} = 0$. Both hold, so the standard filtration argument gives no obstruction at three fields. The resonant $\alpha \in \{1/3, 1/2, 2/3\}$ at three fields and four or more fields were not run.

Theorem 4.4 (F: second implementation)**.** *(i) We compared the Synge data* A, B, C *from the closed Bernoulli form with, independently, the closed exponential form and the explicit angular mode sum* $\sum_n e^{i(n+b)\psi - |n+b|u}$ *with explicit* E_0 *flips. At* $\lambda = 0.04, 0.02, 0.01$ *the residuals scale with order* ≈ 3.00 *for the labels* $0, \alpha, 1-\alpha, 2\alpha, 1-2\alpha$ *(CR) and* $2\alpha - 1$ *(F), as a correct quadratic Synge expansion requires. (ii) We recomputed the spin-2 Ward residual with the mode-sum kernel (*`code/ward_xcheck.py`*, 20 digits):*

α, r	*F, closed form (v1)*	*F, mode sum (v2)*	*CR, mode sum*
0.7071, 1.0	8.61557×10^{-3}	8.61557×10^{-3}	$\leq 8.5 \times 10^{-22}$
0.7071, 0.1	3.71640×10^{-2}	3.71640×10^{-2}	$\leq 2.7 \times 10^{-23}$
0.7071, 0.01	4.48×10^{-2}	4.48472×10^{-2}	$\leq 1.6 \times 10^{-22}$
0.85, 1.0	4.19363×10^{-3}	4.19363×10^{-3}	$\leq 8.5 \times 10^{-22}$

The CR floors are set by the working precision: 10^{-32} *at 30 digits in v1,* 10^{-22} *at 20 digits here. (iii) A third, diagrammatic check of the same dichotomy is Theorem 5.1(iii): the one-loop* $E = 1$ *row is ST-closed on CR and not closed on F.*

5 (C,D) The one-antifield row: one-loop apex poles of D14a,b,c

Method (tadpole/coincidence route). We use the XVI-P graph convention $i\Gamma$ = prefactor $\int \frac{d^d l}{(2\pi)^d} \frac{N}{l^2 + i0}$, the exported vertex tensors $V_{3,\mathrm{grav}}$, $V_{\bar{c}ch}$ and $V_{h\phi\phi}$, and the exported prefactors $-\kappa/2$, κ and $-\kappa/2$ (imported read-only from the copy). In position space, $\int e^{il(x-x')}/(l^2 + i0) = -iG_E$, and a leg momentum becomes $-i\partial$ at that leg's point. On the cone the numerator Π_{cd} is replaced by $\sum_S (\Pi_S N)_{cd} R_{b(S)}$, where Π_S are the helicity projectors built from $u_t, u_\pm = (e_1 \pm ie_2)/\sqrt{2}, u_z$. The flat part is scaleless. We assert that $\sum_S \Pi_S N = \Pi$ in code. Ghost loops use the vector labels $b(\pm 1)$, and the scalar uses $b = 0$. The integrand is $\propto r^{-2}$ and θ-independent (checked), so $\Gamma_{\mathrm{pole}} = (4-d)^{-1} \int_W 2\pi\alpha\, [r^2 f]_{\theta=0}$. The test fields are T_1: $h_{tt} = r^2$; T_2: $h_{zz} = r^2$; T_3: $h_{11} = h_{22} = r^2$; and T_4: $h_{11} = X^2 - r^2/2$, $h_{22} = Y^2 - r^2/2$, $h_{12} = XY$. The gauge mode $\xi_i = x_i r^2$ gives $T_3 + T_4$, so *ST closure* of the $E = 1$ row requires $\Gamma(T_3) + \Gamma(T_4) = 0$. Longitudinal $O(1,1)$ invariance requires $\Gamma(T_1) = -\Gamma(T_2)$. The Euclidean curvatures are $(\bar{R}, Q) = (-4, -2)$ for T_1 and $(4, -4)$ for T_3, with $Q := \bar{R}_{ii} - 2\bar{R}_{ijij}$. We fit $\Gamma_{\mathrm{pole}}/\mathrm{area} = (4-d)^{-1}[A\bar{R} + BQ]$.

Theorem 5.1 ($E = 1$ row on the cone)**.** *Computed in exact sympy arithmetic in α;* `code/cone_tadpole.py`*,* `logs/v2_onepoint_fit.txt`*. (i)* Scalar loop (D14c): $T_1 = -T_2$ *and* $T_3 = -T_4$*, with*

$$A_s = \frac{\kappa(1-\alpha^2)}{288\pi\alpha}, \qquad B_s = -\frac{\kappa(1-\alpha^2)(1+\alpha^2)}{2880\pi\alpha^3}.$$

That is, $(A_s, B_s) = +\frac{\kappa}{16\pi^2}\times$ *Fursaev's* a_2^Σ *coefficients* $\big(\frac{\pi}{18}\frac{1-\alpha^2}{\alpha}, -\frac{\pi}{180}\frac{(1-\alpha^2)(1+\alpha^2)}{\alpha^3}\big)$ *exactly, so the ratio is* $B/A = -(1+\alpha^2)/(10\alpha^2)$*. (ii)* Graviton (D14a) and ghost (D14b) loops, CR: *each separately satisfies* $T_1 = -T_2$ *but violates closure. The violations are* $T_3 + T_4 = \pm\frac{\kappa(\alpha-1)(3\alpha-1)}{4\pi\alpha}$*, and they cancel exactly in the sum. The closed CR total is*

$$A_{g+gh} = -\frac{\kappa(19\alpha^2-20\alpha+5)}{48\pi\alpha}, \qquad B_{g+gh} = -\frac{\kappa(319\alpha^4-240\alpha^3+40\alpha^2+1)}{1440\pi\alpha^3}.$$

In the normalisation of (i), $a_{\bar R}^{g+gh} = -\frac{\pi(19\alpha^2-20\alpha+5)}{3\alpha}$ *and* $a_Q^{g+gh} = -\frac{\pi(319\alpha^4-240\alpha^3+40\alpha^2+1)}{90\alpha^3}$*. Equivalently,*

$$a^{g+gh} = 2\,a^{\text{scalar}} - \frac{4\pi(2\alpha-1)}{9\alpha}\Big[(7\alpha-4)\,\bar R + (4\alpha-1)\,Q\Big].$$

The "2" is the number of physical polarisations. The remainder vanishes at $\alpha = 1/2$ *and tends to* $-\frac{4\pi}{3}(\bar R + Q)$ *as* $\alpha \to 1$*. (iii)* Friedrichs: *the ghost loop is unchanged (its labels coincide), and the graviton loop changes. The F total violates closure by* $T_3 + T_4 = -\frac{\kappa(\alpha-1)(2\alpha-1)(3\alpha-1)}{3\pi\alpha^3} \neq 0$ *for* $1/2 < \alpha < 1$*. Its fitted coefficients,* $A^F = -\frac{\kappa(\alpha-1)(57\alpha^3-27\alpha^2-8\alpha+4)}{144\pi\alpha^3}$ *and* $B^F = -\frac{\kappa(\alpha-1)(319\alpha^3-881\alpha^2+519\alpha-81)}{1440\pi\alpha^3}$*, vanish at* $\alpha = 1$ *and agree with CR at* $\alpha = 1/2$*. They are diagnostic only.*

Reading.

- *The one-antifield row coefficient.* The $E = 1$ ST row is the statement that the one-point apex pole is σ_0-closed, i.e. lies in the 2-dimensional span of $\bar R$ and Q of v1 Proposition 3.4. On CR it does, with the coefficients above, and the one-loop κ^1 apex counterterm is $-\Gamma_{\text{pole}}$. This is the first explicit check that the cone regularisation respects ST at the apex in a sector where graviton and ghost contributions must conspire.

- *F.* The non-closure on F is the one-loop manifestation of the v1 domain obstruction: s does not preserve the F domain for $\alpha > 1/2$.

- *Flat limit.* The nonzero CR flat limit is not a contradiction. At $\alpha = 1$ the CR labels $b(-2) = 1 - 2\alpha = -1$ select the singular E_0 solution in the $(\pm 2, \mp 1)$ channels, so CR at $\alpha = 1$ is not the flat Friedrichs theory; F does vanish at $\alpha = 1$. This parallels FM's spin-2 statement that the conical traces do not tend to the smooth ones as $\beta \to 2\pi$ (FM eq. (2.13), $\bar A^{(2)} = N^{(2)}A^{(0)} + (2(d+2)(\beta-2\pi)+8\pi)\int_\Sigma$; checked in the downloaded text) and the contact-term discussions [K95, S15, BC24]. Those references concern the $O(R^0)$ area coefficient. Our coefficients are at $O(R)$, the a_2^Σ level. We are not aware of a published graviton a_2^Σ for this gauge and boundary condition, so (ii) is a new, unreviewed computation and is not matched to the literature.

Sign convention (ς, resolved). The XVI-P rule $i\Gamma = \text{prefactor}\int N/(l^2 + i0)$, together with $\int e^{il\Delta}/(l^2 + i0) = -iG$, gives $\Gamma = -\text{prefactor}\cdot[\dots]$. Our first code pass (`cone_tadpole.py`) used +prefactor, contradicting its own comment line. This is a common factor $\varsigma = -1$ for all three

graphs. The leg-derivative phase does not affect it, because every term carries two derivatives and $(\pm i)^2 = -1$. All numbers in Theorem 5.1 *include* the correction: they equal $-1\times$ the raw JSON values in `logs/v2_tadpole_*.json`. With the correction, D14c equals $+\frac{\kappa}{16\pi^2} a_2^{\Sigma}$. This agrees with the sign expected from $\Gamma_{\text{pole}} = +\frac{1}{16\pi^2(4-d)} \int a_2$ for $g = \eta + \kappa h$, but that cross-check depends on the signature and curvature conventions, so the decisive argument is the XVI-P rule. Closure statements, ratios and the normalised a-coefficients do not depend on ς.

6 (B) The two-point sector with the apex

Theorem 6.1 (B: Ward-constrained values, conditional)**.** *Assume (H_{reg}): the cone regularisation (cone-DRED on CR) is BRST-invariant up to evanescent operators, so that at one loop the apex pole part $\Gamma^{(1)}_{2,\text{pole}}|_W$ satisfies the linearised ST identity. Then its projection onto the 34-dimensional inventory lies in Z ($\dim 8$). In particular the 26 Ward-constrained combinations ($34 - \dim Z$) have pole coefficient 0 for all α off the resonant set, with $42 - 9 = 33$ at $\alpha = 1/2$ and $35 - 8 = 27$ at $\alpha \in \{1/3, 2/3\}$.*

Proof. The ST pole condition is $\sigma_0 \Gamma_{\text{pole}} = 0$, i.e. $\Gamma_{\text{pole}} \in Z$, and the 26 combinations are coordinates on the complement of Z. Evanescent breaking multiplies poles by $(d-4)$ and is finite at one loop, so it affects only finite parts. Any finite apex breaking is WZ-closed and, by Theorem 4.1, exact; it is absorbed by finite inventory counterterms. □

Status of ($\mathbf{H}_{\text{reg}}$). It has three ingredients.

- CR preserves the domain: $K_2 c$ has no foreign atoms in any sample (proved in the engine).
- The vertices are the flat XVI-P ones.
- The $E = 1$ row is closed at one loop on CR and fails on F (Theorem 5.1). This is a nontrivial, sector-restricted confirmation.

We have *not* proved (H_{reg}) for the two-point bubbles. Hence the theorem is conditional.

What remains. The 8 closed directions carry the physical apex data. The 6 exact ones are fixed in value by the graphs but removable by canonical transformations on W, so they are scheme data. The 2 $\phi\phi$ ones are free and physical. Their values require the apex poles of the bubbles below, which are open. The finite parts ($H \cdot R$, $R \cdot R$) are open.

Evanescent apex directions. In DRED the longitudinal momenta are $(d-2)$-dimensional and the vertex indices are quasi-4. Apex candidates involve $p_\parallel$ only through $O(d-2)$ invariants, so every candidate with a $p_\parallel^2$ factor has an evanescent twin with $\hat{p}^2$. These arise from the jet-pair classes with $k = 2$: three in hh, one in $\bar{c}c$ and one in $\phi\phi$; the Kc classes have $k \leq 1$. Their one-loop coefficients are finite (pole $\times(d-4)$) and are fixed by the same open bubbles. In the $E = 1$ row there are none: the one-point candidates have $k = 0$, and we verified that the $k = 1$ terms are total derivatives on W.

7 (C) D01–D12 analogues on the cone

Graph (XVI-P/Q)	Flat status (read-only sources)	Cone apex pole/log part
D14a/b/c ($E = 1$)	scaleless, 0	**computed** (Thm. 5.1)
D01 hh graviton bubble	part of Q02	open: bubble
D03 ghost bubble	part of Q02	open: bubble
D06 (=D01+D03 channel)	part of Q02	open: bubble
D04 mixed $h\phi$ bubble	0 (DRED: $VPV' = 2p^2(l-p)^2$, scaleless; Q02_D04_scrutiny)	open: the cancellation needs the full Π, but on the cone each helicity sector has its own $R_{b(S)}$, so the flat identity does not reduce D04 to a tadpole
D11 scalar bubble	inherited XVI-O kernel	open: bubble
D13 Kc ghost–graviton	part of Q02	open: bubble
D02 h^4, D12 $hh\phi\phi$, D05a tadpoles	scaleless	computable by the route of §5; not done (projection of two-leg coincidence data onto the 18-dimensional hh jet basis not implemented)

Precise obstruction for the bubbles. At an apex the pole of a two-propagator graph comes from the region where $x, x' \to W$ together. There $G = H + R$ with $R \sim (rr')^{-1}\Phi_b(\eta, \psi)$, and the products $H \cdot R$ and $R \cdot R$ are not determined by Synge coincidence data, because R is not smooth at $r = r' = 0$. One needs the apex-supported renormalised product of these distributions, with overlapping bulk ($x = x'$) and apex ($r, r' \to 0$) divergences, for example via a Kontorovich–Lebedev/Mellin representation in (r, r'). This is not available here. The pole *structure* is constrained by Theorem 6.1. The *values* are open.

8 Corrections to v1

1. *Jet cap.* The v1 engine built jets up to $T - 1$ in `candidates`. This can miss partners of CR-flipped jets with $m \in (-1, 0)$. v2 uses cap T and re-ran every α: no count changed, so all v1 claims stand. The v1 engine is preserved as `apex_jets_v1_frozen.py`.

2. *Nature of the ranks.* The v1 wording "exact modular arithmetic" is sharpened: the ranks are probabilistic Schwartz–Zippel certificates at two primes.

3. *Resonance $\alpha = 1/3$.* v1 listed it as predicted and not run. It is now run (CR: $N = 35$, $H^0 = 2$, $H^1 = 0$). v1 also listed $1/2$ and $2/3$ as the resonances; the structure theorem shows $\{1/3, 1/2, 2/3\}$ is the complete set.

4. *Scope of the FM remark.* v1 quoted Fursaev's a_2^Σ conditionally for the scalar. It is now computed, and the match includes the normalisation $-\kappa/16\pi^2$.

9 Literature

Markers: * verified online with the cited equation checked in the downloaded text; ‡ verified online at bibliography/abstract level; † not verified.

* [FM97] D. V. Fursaev, G. Miele, NPB 484 (1997) 697, hep-th/9605153: eqs. (2.11), (2.13), (3.40), (3.47); spin-2 traces do not reduce to smooth ones as $\beta \to 2\pi$.

* [S11] S. N. Solodukhin, Living Rev. Rel. 14 (2011) 8, arXiv:1104.3712: eq. (70), the a_2^{Σ} used in Thm. 5.1(i) (v1 check).

‡ [S15] S. N. Solodukhin, "Newton constant, contact terms and entropy", PRD 91 (2015) 084028, arXiv:1502.03758: graviton contact term $c^{(2)}(d) = (d^2 - d + 4)/2$; non-analytic spin-2 terms; orbifold vs. n-fold cover.

‡ [BC24] A. Blommaert, S. Colin-Ellerin, "Gravitons on the edge", arXiv:2405.12276 (tip removed, regular modes): Kabat's method for gravitons; a graviton contact term equal to $(d-2)$ times Kabat's photon term.

‡ [K95] D. Kabat, NPB 453 (1995) 281, hep-th/9503016.

‡ [FS95] D. V. Fursaev, S. N. Solodukhin, PRD 52 (1995) 2133. [F94] D. V. Fursaev, PLB 334 (1994) 53.

‡ [D77] J. S. Dowker, J. Phys. A 10 (1977) 115. [LW96] F. Larsen, F. Wilczek, NPB 458 (1996) 249.

All of these concern heat-kernel coefficients at the area (R^0) level, or scalars and vectors at $O(R)$. None provides the graviton+ghost $O(R)$ apex coefficients of Theorem 5.1(ii) in de Donder gauge with CR. Citations are used for comparison only. None is promoted to a theorem.

10 Reproducibility

The scripts are in `/workspace/XVI-O1-CONE/code` and the logs in `logs/`:

- `flat_count.py`, `resonances.py`, `batch_v2.sh`/`apex_run.py` (`logs/v2apex/`);
- `coinc.py`, `xcheck_coinc.py`, `ward_xcheck.py`;
- `cone_tadpole.py` (D14a/b/c, CR and F);
- `tri_count.py`, `tri_run.py` (`logs/v2_tri_*.json`).

The second trilinear run at $\alpha = 309/500$ is recorded in `CHECKPOINT_v2.md`. The XVI-P, XVI-Q and XVI-R tree hashes (`logs/frozen_tree_*.sha`) are identical before and after. Nothing was pushed or written to the site.

11 Ledger

Item	Status
Regular-region locality	proved (v1)
FM / Kabat coefficient matches	computed, exact (v1)
A1 flat count 5/2/3/1 in the apex engine	computed
A2 resonances 1/3 (new), 1/2, 2/3; 1/4, 2/5 non-resonant	computed (2 primes, both branches)
A3 structure theorem; $H^0_{(2)} = 2$, $H^1_{(2)} = 0$ for all α (CR)	proved modulo Schwartz–Zippel
B 26 Ward-constrained pole values = 0	proved conditional on ($\mathrm{H_{reg}}$)
B 6 exact + 2 $\phi\phi$ coefficients; finite parts	open (bubbles)
C/D $E = 1$ row: D14a,b,c apex poles, closure on CR	computed (exact in α)
C/D Scalar = Fursaev a_2^Σ incl. normalisation	computed
C/D F-branch $E = 1$ row	not closed (obstruction; diagnostic)
C D01, D03, D04, D06, D11, D13 apex poles	open (renormalised apex products)
C D02, D12, D05a apex poles	open (computable; not implemented)
C evanescent apex directions	identified; coefficients open
E $H^1_{(3)} = 0$ generic α, CR	computed (2 primes; 2 α)
E resonant three-field; ≥ 4 fields; full nonlinear descent	open
F mode-sum cross-checks (Synge data, Ward residuals)	computed
Overall sign ς of D14	resolved ($\varsigma = -1$ from the XVI-P rule; applied)
Independent review	not started (not requested)
O.1 on the cone	**NOT_CLOSED**
O.1 (flat, scoped Q11)	CLOSED (scoped), untouched
`bulk_symmetry_restored`	false

Why NOT_CLOSED. A scoped analogue of Q11 needs every two-point graph's apex pole, projected and shown to satisfy ST. The bubbles are open, and ($\mathrm{H_{reg}}$) is unproved for them. A CLOSED_PENDING_INDEPENDENT_REVIEW status would require both, so it is not claimed for any scope.

Non-claims

We claim no Einstein dynamics and no emergence. The cone and the CR extension are inputs. No separation is used as a bridge. `bulk_symmetry_restored` = `false`. Q11, XVI-P, XVI-Q and XVI-R are unmodified. A missing result is never counted as zero, and partial results are reported as partial. The graviton loop coefficients are new and unreviewed.